Spatio-Temporal DeepFake Detection using Vision Transformers with Temporal Attention

1st Sai Sreenadh Chilukuri*, 2nd Mallipeddi Harsha Vardhan†, 3rd Sameer Zyad ‡
School of Computer Science and Engineering, Vellore Institute of Technology
Andhra Pradesh, India
Email: *sreenadh.23bce7616@vitapstudent.ac.in, †harsha.23bce8869@vitapstudent.ac.in, ‡sameer.23bce8876@vitapstudent.ac.in

***Abstract*—In this study, a deepfake detection system is presented that utilizes Deep learning models to detect manipulated videos. Deepfake technology has many uses, including security systems, entertainment, and educational experiences. However, because deepfake technology has become easier to access than ever before, it is being exploited for unethical and illegal purposes. Therefore, the development of deepfake detection models is crucial to mitigate this issue. We propose a vision transformer and temporal transformer model trained on the FaceForensics++ dataset to identify inconsistencies in video footage by detecting anomalous patterns in frame data. This study explores the improvements in this model's accuracy compared to other Deep Learning models with better architectures, as well as the importance of complex systems in mitigating the negative effects of deepfake technology. This model achieved an accuracy of 94% in video data.**

***Index Terms*—Deepfake Detection, Vision Transformer, CNN, LSTM, Deep Learning**

I. Introduction

Deepfake technology is one of the most common and available forms of artificial intelligence-based technology. The term "Deepfake" comes from the words "Deep learning" and "fake". Deepfake technology uses generative adversarial networks, consisting of a pair of neural networks: one that generates the deepfake and the other that detects and refines the deepfake content. However, this technology is mostly abused. for unethical and illegal purposes, such as defamation, identity theft, fraud, pornography, and misinformation. The rapid development of AI and deepfake technology only serves to exacerbate the problem. Fig. 1 shows an example of a manipulated frame of an American right-wing political activist Charlie Kirk's face swapped onto a video of the Twitch streamer Ishowspeed using deepfake technology. Many high-profile members of the public, such as politicians, celebrities, activists, and entrepreneurs, have had their careers jeopardized by deepfakes. Such figures include Barack Obama, Charlie Kirk, Kamala Harris, and Joe Biden. Since 2017, approximately 99-100% of the victims of deepfake pornography victims have been women [1]. Although deepfakes can be useful in the education sector as tools for learning to reduce human workload, the potential for misuse and the ease with which they can be created pose severe issues in every aspect of the digital world. Deepfake generation of human faces is divided into four categories: re-enactment, replacement, editing, and synthesis [2]. These categories are then used by neural networks, such as Convolutional Neural Networks, Generative Adversarial Networks, and Recurrent Neural Networks (RNN). Figure 2 depicts the exponential growth of Deepfake usage in media from 2023 to 2025 Q1 [1]. While creating deepfakes, some inconsistencies are left behind, which can be used to determine whether they are fake. Models such as XceptionNet [3] and FakeSpotter [4] are widely used for deepfake detection.

Fig. 1: Deepfake Image

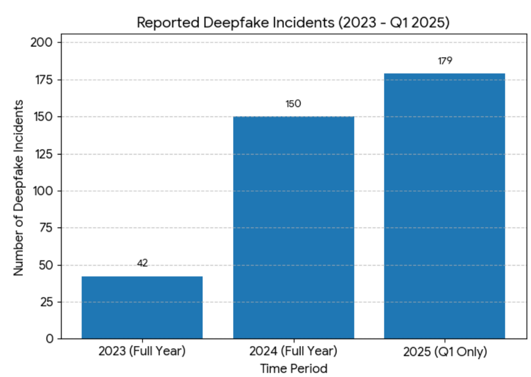


Fig. 2: No of deepfake cases reported. Information source [1]

II. Literature Review

Deepfakes threaten the integrity, reliability, and privacy of information on the internet. The field of deepfake technology has attracted significant attention, and multiple deepfake detection models have been created as a result. Deepfake detection

Fig. 3: Different deepfake algorithms in FaceForensics++ dataset [5]

methods can be classified into statistical, machine learning, and deep learning models [6]. Several benchmark models perform different processes, such as XceptionNet [3], which is more efficient and captures specific details and intricate patterns in the image. A study by Khatri et al. (2023) [7] compares models such as ResNet, which can train very deep neural networks effectively, CNNs [8], which have robust architectures and high accuracy, along with XceptionNet [3]. Another benchmark model is CapsuleNet [9], which captures hierarchical spatial relationships between facial features. This model is highly robust and accurate and utilizes the faceForensics++ [5] dataset. For efficiency, MesoNet [10] provides fast real-time responses to image inputs, but struggles with high-quality deepfakes. Most models work well on training datasets, but their accuracy declines sharply on unseen data due to overfitting. Additionally, CNN-based [8] methods such as XceptionNet [3] and MesoNet [10] rely on pixel-level feature detection, which results in decreased accuracy on compressed or noisy data. Pure CNN [8] models have limited temporal understanding, i.e., they fail to consider motion continuity, which affects accuracy. To solve these shortcomings, Vision transformers [11] are utilised. Vision transformers are robust, exhibit strong generalisation, at capturing spatial features, and offer better Interpretability through attention maps.

III. Methodology

There are three main techniques used for deepfake detection and creation: Attribute-based detection, feature-based detection, and data-driven models [12]. The model we are proposing is a data-driven model. Our proposed model is trained on the FaceForensics++ dataset [5], which contains both real and fake videos generated through multiple deepfake algorithms. Fig. 5 illustrates the workflow of the model.

A. Data Collection

Our proposed model was trained and tested on FaceForensics++ dataset [5], which consists of more than 1000 real and deepfake videos with different amounts of compression and added noise, to test the robustness of the model. Videos are sourced from youtube, and are selected so that they feature mostly front-facing, trackable faces. it contains a total of 5000 videos (1000 real + 6000 fake). The with over 1.8 million frames in total for all compression levels. FaceForensics++ [5] features six main manipulation methods: Facial Expression Manipulation (Reenactment) and Facial Face swapping (identity manipulation).

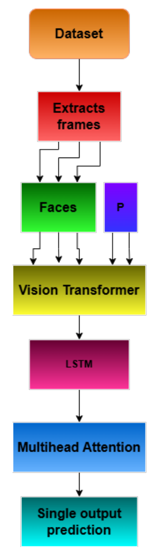


Fig. 5: Methodology diagram of the proposed model

- **Face2Face [13]**: a computer graphics-based method to transfer the source's facial features to the target in real time.
- **FaceSwap [14]**: a computer graphics-based approach for replacing the face of a person with the face of another person.
- **DeepFakes [15]**: a popular deep-learning method for face swapping.
- **NeuralTextures [16]**: a deep-learning-based method using Generative Adversarial Networks.
- **FaceShifter [17]**: a two-stage face swapping method able to generate high fidelity identity preserving face swap results.

This trains the model on several sets of visual conditions. The real and deepfake content are balanced to prevent bias in training. For our proposed model we train our dataset on the FaceSwap section of the dataset.

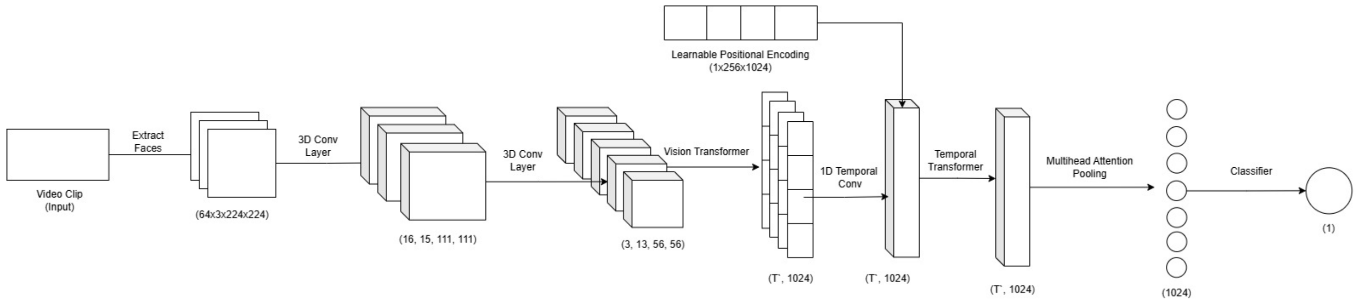


Fig. 4: Proposed model architecture diagram

B. Preprocessing

This is slightly different from the traditional preprocessing step for CNN [8] architecture, where explicit feature extraction, such as face detection or cropping, takes place. A ViT [11] views an image as a sequence of fixed-size tokenised patches. Preprocessing first resizes every frame to 224 × 224 pixels, matching the pretrained model parameters. The pixels are then normalised between 0 and 1 or standardised based on the ImageNet [18] mean and standard deviation to stabilise learning. An image is divided into 16 × 16 non-overlapping patches, creating 196 segments from a 224 × 224 input. Each patch is flattened and goes through a linear projection layer, resulting in dense embeddings as visual tokens. Since Transformers inherently hold no spatial awareness, positional encodings are added to store patch location information. Also, a [CLS] token is prepended for the final classification. Feature augmentations, including flipping, colour jittering, and cropping, can take place before patching as optional preprocessing steps in order to increase generalisation. Positional contextualization of the resulting patch embeddings allows them to be input into the Transformer encoder. The multihead self-attention mechanism in the Transformer then identifies subtle manipulations across the image to perform the task of classifying frames as real or fake.

C. Architecture

The deep learning model used in our work is a vision transformer [11] with temporal attention. This model architecture can be broadly divided into layers as follows:. Figure 4:

- **Backbone**: The backbone of the model comprises a Vision Transformer [11], which acts as a feature extractor for each video frame. The model uses the VIT_L_16 [11] architecture from the torchvision library, pretrained on ImageNet-1k, ensuring strong generalisation of visual patterns. The transformer processes each frame individually to produce global spatial relationships. The classification head is replaced with an nn.identity layer so that only feature representation is extracted from the frames. Additionally, a feature dimension map determines the correct embedding size for each model variant. The parameters are initially frozen to keep the weights constant during early training. Later, selected layers are unfrozen to enable fine-tuning of higher-level features. Temporal Transformer models long-range dependencies across video frames. It helps depict how visual features, for example, facial expressions, morph over longer time scales. It consists of transformer encoder [19] layers that are stacked over each other; each layer is made up of multi-head self-attention. This layer serves to encode the motion patterns from an input sequence. Every frame is allowed to interact with all other frames, helping it learn global temporal relationships.
- **Temporal Stem**: A temporal stem is a 3D Convolutional block that is designed to capture continuity across consecutive frames. It takes input in the form of (B, C, T, H, W) and processes it using two 3D convolutional layers with batch normalisation. The first layer expands from 3 to 16 channels and reduces temporal dimensions, while the second layer reduces the channels back to 3. It is done because ViT supports 3 channels. Thus, spatiotemporal information is encoded with low computational costs.
- **Positional Encoding**: Positional encoding is a method that provides information about the order in which the model should consider the frames of a video. This is because, on their own, transformers do not have any temporal awareness. It relies on a tensor in which each frame is assigned a frame index positionally. These encodings are added to the feature vector so that the model knows the position of each frame in the sequence. For shorter videos, only part of the encoding is used. In this way, a fixed-size temporal context is preserved, no matter the length of the video.
- **Multihead Attention Pooling**: The pooling layer combines information from all the frames into a single video-level representation. It uses multihead attention pooling, allowing the model to learn which frames are most important for classification. Attention weights are assigned to each frame depending on the importance it contributed to the final prediction. Every head calculates attention scores highlighting different temporal cues such as facial distortions or irregular movement. These are then aggregated to produce a context vector representing the most informative moments. This is fused with a mean-

pooled representation using a weighted sum:

$$weighted\ sum = 0.6 \times attention\ context + 0.4 \times mean\ context \quad (1)$$

This ensures a focused and balanced temporal understanding.

- **Classifier**: The classifier, or classification head, converts the spatiotemporal features extracted thus far and converts them into prediction scores. It first applies LayerNorm to stabilise the feature distribution, then dropout to reduce overfitting. The final layer maps the normalised feature vector to the number of output categories the model predicts. This produces a single output value per video —a clear, interpretable indicator of whether the video is real or fake.

IV. Results

We evaluate our model on standard deepfake benchmarks to demonstrate the effectiveness of temporal reasoning in enhancing detection accuracy. To assess the effectiveness of our proposed architecture (Figure 4), extensive experiments were conducted on the FaceForensics++ dataset [5]. To identify the best-performing model, we evaluate both quantitative and qualitative aspects. Under quantitative metrics, the metrics we consider are:

- **Accuracy**: The proportion of the total correct predictions.
- **Area under the curve(AUC)**: The probability that the model will rank a randomly chosen positive instance higher than a randomly chosen negative instance.
- **F1-Score**: The harmonic mean of precision and recall. It balances the cost of false positives and false negatives.

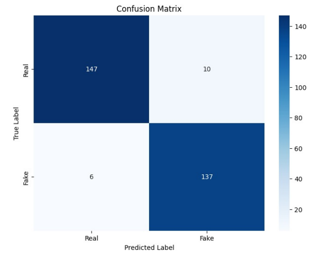
We test vision transformers paired with different temporal models to assess the impact of improved temporal features on deepfake detection. From the FaceForencics++ FaceSwap dataset, we use 15% of videos for validation and 15% for testing. When testing our architecture with a Bi-LSTM [20] and a Transformer, we achieved better performance by utilising the temporal transformer. See Table II for the metrics observed with different architectures.

TABLE I: Our Proposed Model Results

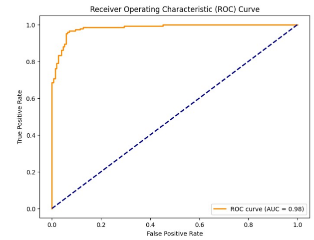
Deepfake algorithm	**Performance Metrics**		
	Accuracy	***AUC***	***F1-Score***
FaceSwap	94.67%	98.31%	95%

As shown in Table I, our model achieves an accuracy of 94.6%, AUC of 98.31% and a F1-score of 95%, outperforming previous spatial-based deepfake detection models. The high AUC reflects the model's strong capability to distinguish between authentic and manipulated videos across varying thresholds.

In addition to numerical data, we show the training behaviour and classification ability of our model. The loss curve (Figure 7) shows steady convergence with little overfitting. This suggests effective generalisation. The confusion matrix



(a) Confusion Matrix



(b) ROC Curve

Fig. 6: Metrics

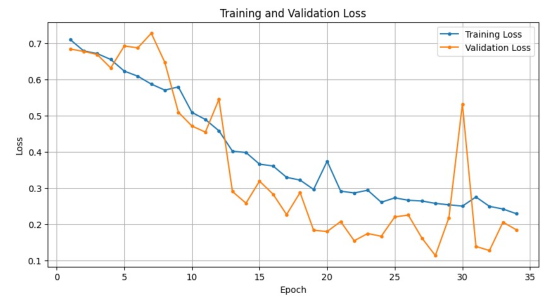


Fig. 7: Training and validation loss vs epoch

(Figure 6a) displays a strong diagonal dominance, which indicates good separation between real and fake samples. Moreover, the ROC curve (Figure 6b) demonstrates a high true positive rate and a large area under the curve. This supports the model's strength in detecting subtle manipulations.

TABLE II: Performance Metrics of Different Deepfake Detection Architectures on the FaceSwap Dataset

Architecture	**Accuracy (%)**	**AUC (%)**	**F1-Score (%)**
CNN + Mean Pooling	86.67%	94.68%	87.00%
ViT_B_16 + BiLSTM [20]	88.67	95.14	88.00
ViT_L_16 + BiLSTM [20]	92.00	97.01	91.00
ViT_B_16 + Temporal Stem + BiLSTM [20]	92.33	97.79	92.00
Proposed Model	**94.67**	**98.31**	**95.00**

The proposed model shows a significant improvement compared to traditional CNN-based deepfake detection methods. While CNNs are limited in capturing long-range dependencies and temporal coherencies, our hybrid ViT based framework combines a pre-trained Vision Transformer backbone with a 3D temporal stem and multihead attention fusion to competently catch subtle inter-frame inconsistencies and fine-grained spatial-temporal correlations. Empirically, this outperforms prior CNN methods with an accuracy of 94.67%, AUC of 98.31%, and F1-score of 95.00%, establishing a new benchmark in performance for temporal deepfake detection.

V. Conclusion

On the whole, deepfake technology is used in social and digital media to defame a person using manipulated media with the person's likeness. This technology has significant potential to harm people, and sufficient measures must be taken to mitigate that harm. The threat posed by deepfake technology is versatile and multi-dimensional; this is a relatively new field of study, so there is great potential regarding threats that could be posed by this technology. The work proposed here is based on the deep learning model trained on vision transformers and the FaceForensics++ dataset [5]. Our model has been able to achieve 94% accuracy in manipulated video detection, higher than most models.

Deepfakes have quickly emerged as one of the most ominous digital threats in this modern age because their misuse can diffuse misinformation, destroy reputations, and manipulate public opinion. On social media, this fabricated video can spread virally, which can have serious social, political, and psychological impacts. Thus, realistic mimicry of real people creates an urgent challenge for developing advanced detection systems that differentiate between authentic and synthetic videos with high reliability. In spite of numerous studies on CNN [8] and RNN-based models, most of them still remain capable of poor generalisation across different datasets and types of manipulations.

Our proposed transformer-based model overcomes these limitations by learning effective spatial and temporal dependencies in video data, improving robustness and generalisation across unseen manipulations. Trained on the high-quality FaceForensics++ dataset, this model learns to find minute inconsistencies in facial motions, lighting, and texture patterns that are well beyond human-visual detectability. The system achieves an accuracy of 94% and demonstrates the potential for transformer-based architectures in the development of more secure and trustworthy media verification tools. Continued research and refinement in the area will be required to harden digital defences and protect the integrity of online content into the future.

VI. Future work

In the near future, the generation models of deepfakes will be refined, making the task of distinguishing between deepfakes and original videos increasingly difficult. We believe that identifying the weaknesses and limitations in existing deepfake detectors, such as the limitations on vision transformers [11], could form part of the future work in this context. Only when researchers are aware of these vulnerabilities will they be able to develop stronger countermeasures against them.

While methods for creating deepfakes continue to improve through the development of more sophisticated architectures, coupled with better data synthesis techniques, the traditional approaches to detection can no longer ensure the reliability and accuracy of a model. Future work should hence be directed toward not only improving the accuracy of classification but also toward analysing why certain detection systems fail under specific conditions, such as compression or lighting variation, or unseen manipulation techniques. Secondly, large-scale cross-dataset evaluations need to be conducted in order to see how well these models generalise outside controlled datasets.

A closer look into the internal structure of models such as vision transformers, by studying attention mechanisms [19], feature extraction processes, and sensitivity to noise, can uncover useful knowledge about their limits of operation. Knowing these elements will enable the research community to devise more explainable, adaptive, and robust systems that are able to detect forgeries that are increasingly realistic. Eventually, having covered most of the shortcomings, while also considering future evolution in deepfake generation, researchers will be in a better position to construct detection systems that can maintain digital integrity and trust in a world where synthetic media will continue to progress rapidly.

References

[1] Keepnet Labs, "Deepfake statistics & trends 2025: Growth, risks, and future insights," https://keepnetlabs.com/blog/deepfake-statistics-and-trends, September 2025, accessed: 2025-11-10. [Online]. Available: https://keepnetlabs.com/blog/deepfake-statistics-and-trends

[2] Y. Mirsky and W. Lee, "The creation and detection of deepfakes: A survey," *ACM computing surveys (CSUR)*, vol. 54, no. 1, pp. 1–41, 2021.

[3] I. Kusniadi and A. Setyanto, "Fake video detection using modified xceptionnet," in *2021 4th International Conference on Information and Communications Technology (ICOIACT)*. IEEE, 2021, pp. 104–107.

[4] R. Wang, F. Juefei-Xu, L. Ma, X. Xie, Y. Huang, J. Wang, and Y. Liu, "Fakespotter: A simple yet robust baseline for spotting ai-synthesized fake faces," *arXiv preprint arXiv:1909.06122*, 2019.

[5] A. Rossler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "Faceforensics++: Learning to detect manipulated facial images," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 1–11.

[6] M. S. Rana, M. N. Nobi, B. Murali, and A. H. Sung, "Deepfake detection: A systematic literature review," *IEEE access*, vol. 10, pp. 25 494–25 513, 2022.

[7] N. Khatri, V. Borar, and R. Garg, "A comparative study: deepfake detection using deep-learning," in *2023 13th International Conference on Cloud Computing, Data Science & Engineering (Confluence)*. IEEE, 2023, pp. 1–5.

[8] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 2002.

[9] N. Van Quang, J. Chun, and T. Tokuyama, "Capsulenet for micro-expression recognition," in *2019 14th IEEE international conference on automatic face & gesture recognition (FG 2019)*. IEEE, 2019, pp. 1–7.

[10] D. Afchar, V. Nozick, J. Yamagishi, and I. Echizen, "Mesonet: a compact facial video forgery detection network," in *2018 IEEE international workshop on information forensics and security (WIFS)*. IEEE, 2018, pp. 1–7.

[11] A. Dosovitskiy, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020.

[12] Y. Patel, S. Tanwar, R. Gupta, P. Bhattacharya, I. E. Davidson, R. Nyameko, S. Aluvala, and V. Vimal, "Deepfake generation and detection: Case study and challenges," *IEEE Access*, vol. 11, pp. 143 296–143 323, 2023.

[13] J. Thies, M. Zollhofer, M. Stamminger, C. Theobalt, and M. Nießner, "Face2face: Real-time face capture and reenactment of rgb videos," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 2387–2395.

[14] M. Kowalski, "FaceSwap," GitHub, 2017, accessed: Nov. 11, 2025. [Online]. Available: https://github.com/MarekKowalski/FaceSwap/

[15] deepfakes and contributors, "Deepfakes," GitHub, 2024, accessed: Nov. 11, 2025. [Online]. Available: https://github.com/deepfakes/faceswap

[16] J. Thies, M. Zollhöfer, and M. Nießner, "Deferred neural rendering: Image synthesis using neural textures," *Acm Transactions on Graphics (TOG)*, vol. 38, no. 4, pp. 1–12, 2019.

[17] L. Li, J. Bao, H. Yang, D. Chen, and F. Wen, "Faceshifter: Towards high fidelity and occlusion aware face swapping," *arXiv preprint arXiv:1912.13457*, 2019.

[18] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "Imagenet: A large-scale hierarchical image database," in *2009 IEEE conference on computer vision and pattern recognition*. Ieee, 2009, pp. 248–255.

[19] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in neural information processing systems*, vol. 30, 2017.

[20] A. Graves and J. Schmidhuber, "Framewise phoneme classification with bidirectional lstm and other neural network architectures," *Neural networks*, vol. 18, no. 5-6, pp. 602–610, 2005.